Grade 9 Mathematics – Axioms

Answer Sheet – Worked Solutions

Chalaka Perera | BSc (Hons) Computer Engineering, University of Peradeniya

Axioms are basic facts accepted as true without proof. The axioms used in this lesson are:

1. Quantities which are equal to the same quantity are equal to one another.
2. If equals are added to equals, the sums are equal.
3. If equals are subtracted from equals, the remainders are equal.
4. If equals are multiplied by equals, the products are equal.
5. If equals are divided by equals (not zero), the quotients are equal.
6. The whole is greater than any of its parts.

Question 1

(i) *(equal to the same quantity)* (ii) *(equals added to equals)* (iii) *(equals subtracted from equals)*

(iv) *(equals multiplied by equals)* (v) *(equals divided by equals)* (vi) *(the whole is greater than the part)*

Question 2

(i) a = **c** *(equal to the same quantity)*

(ii) x – z = **y – z** *(equals subtracted from equals)*

(iii) p = **q** *(equals divided by equals)*

Question 3

(i) x + 7 – 7 = 12 – 7 *(equals subtracted from equals)* → **x = 5**

(ii) x/3 × 3 = 4 × 3 *(equals multiplied by equals)* → **x = 12**

(iii) 5x – 4 + 4 = 21 + 4 *(equals added to equals)* → 5x = 25; 5x ÷ 5 = 25 ÷ 5 *(equals divided by equals)* → **x = 5**

(iv) 2(x + 3) ÷ 2 = 18 ÷ 2 *(equals divided by equals)* → x + 3 = 9; x + 3 – 3 = 9 – 3 *(equals subtracted from equals)* → **x = 6**

Question 4

AB = CD (given). Add BC to both sides: AB + BC = CD + BC *(equals added to equals)*.

But AB + BC = AC and BC + CD = BD. Therefore **AC = BD**.

Question 5

PR = QS (given), that is PQ + QR = QR + RS.

Subtract QR from both sides: PQ + QR – QR = QR + RS – QR *(equals subtracted from equals)*. Therefore **PQ = RS**.

Question 6

AÔB = CÔD (given). Add BÔC to both sides: AÔB + BÔC = CÔD + BÔC *(equals added to equals)*.

But AÔB + BÔC = AÔC and BÔC + CÔD = BÔD. Therefore **AÔC = BÔD**.

Question 7

SQ̂R = ½ PQ̂R and TR̂Q = ½ PR̂Q (QS and RT are bisectors).

Since PQ̂R = PR̂Q, dividing both by 2: ½ PQ̂R = ½ PR̂Q *(equals divided by equals)*. Therefore **SQ̂R = TR̂Q**.

Question 8

(i) *(equals added to equals)* (ii) *(equals divided by equals)* (iii) *(equals multiplied by equals)*

Question 9

(i) AB = AC and AC = BC, so AB and BC are both equal to AC. Therefore **AB = BC** *(equal to the same quantity)*.

(ii) AB = BC = AC, so ABC is an **equilateral triangle**.

(iii) AC = AB + BC, so AB is a part of AC. Therefore **AC > AB** *(the whole is greater than the part)*.

Question 10

y + 4 – 4 = 10 – 4 *(equals subtracted from equals)* → y = 6.

x = y and y = 6, so x and 6 are both equal to y. Therefore **x = 6** *(equal to the same quantity)*.

Question 11

(i) *(equals divided by equals)*

(ii) This axiom can only be used when we divide by a **non-zero** quantity. Division by zero is not defined, so we cannot divide both sides by 0. Therefore the conclusion 5 = 3 is wrong.